

Ms Lau
Mrs Kerr

Name:

Teacher:.....



Pymble Ladies' College

**HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION
2016**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen.
Black pen is preferred.
- Board approved calculators may be used.
- A reference sheet is provided.
- In Questions 11-16, show relevant mathematical reasoning and/or calculations.
- Start each question in a new booklet.

Total Marks – 100

Section I Pages 1-6

10 marks

- Attempt all Questions 1-10
- Allow about 15 mins for this section

Section II Pages 7-15

90 marks

- Attempt Questions 11-16
- Allow about 2 hour 45 minutes for this section

Mark	/100
Highest Mark	/100
Rank	

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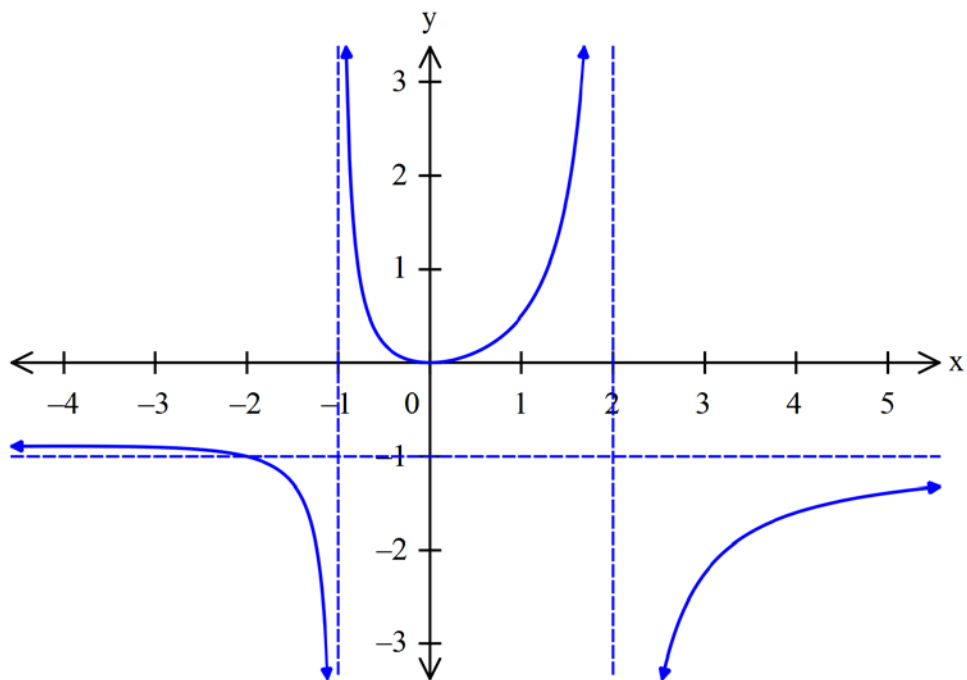
Section I

10 marks

Attempt Questions 1-10

Use the multiple choice answer sheet for Questions 1-10.

- 1 The graph of the function $y = \frac{g(x)}{(x-2)(x+1)}$ is shown below.



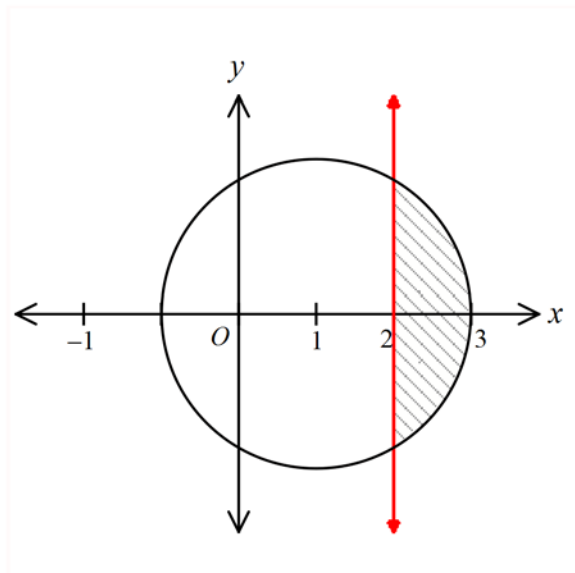
The function $g(x)$ could be which of the following?

- (A) x
 - (B) $-x$
 - (C) x^2
 - (D) $-x^2$
- 2 What type of conic section is represented by the equation $y^2 - 4y - x + 3 = 0$?
- (A) Hyperbola
 - (B) Circle
 - (C) Parabola
 - (D) Ellipse

3 What are the values of real numbers p and q such that $1-i$ is a root of the equation $z^3 + pz + q = 0$?

- (A) $p = 2$ and $q = 4$.
- (B) $p = 2$ and $q = -4$.
- (C) $p = -2$ and $q = 4$.
- (D) $p = -2$ and $q = -4$.

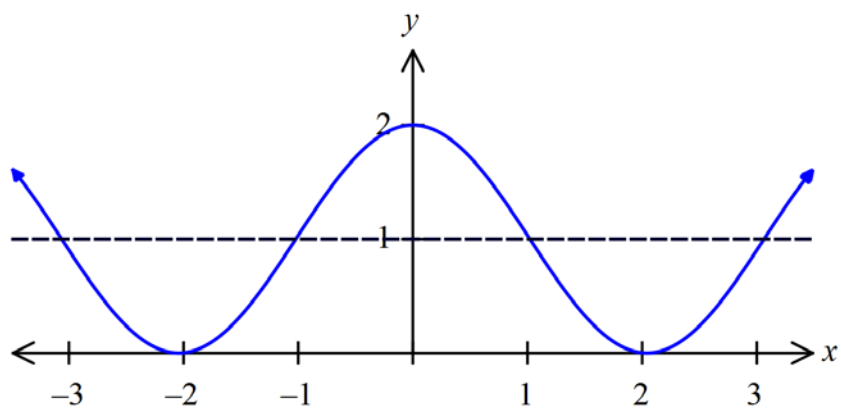
4 Consider the Argand diagram below.



Which inequality could define the shaded area?

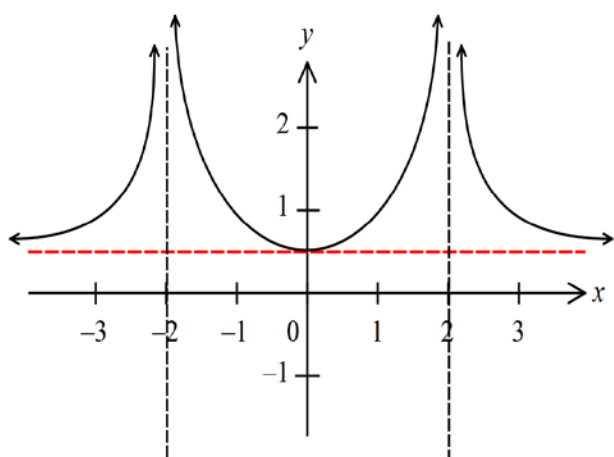
- (A) $|z-1| \leq 2$ and $\text{Re}(z) \geq 2$.
- (B) $|z-1| \leq 2$ and $\text{Im}(z) \geq 2$.
- (C) $|z+1| \leq 2$ and $\text{Re}(z) \geq 2$.
- (D) $|z+1| \leq 2$ and $\text{Im}(z) \geq 2$.

- 5 The graph of the function $y = f(x)$ is shown below.

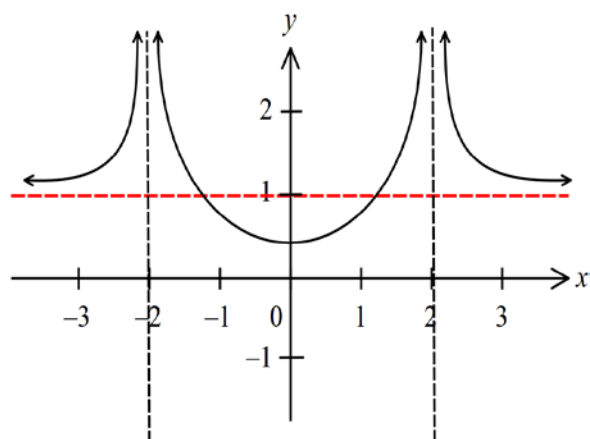


Which graph shows the most accurate representation of the function $y = \frac{1}{f(x)}$?

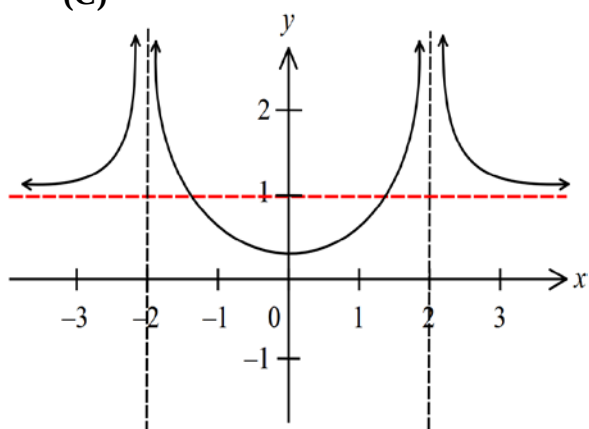
(A)



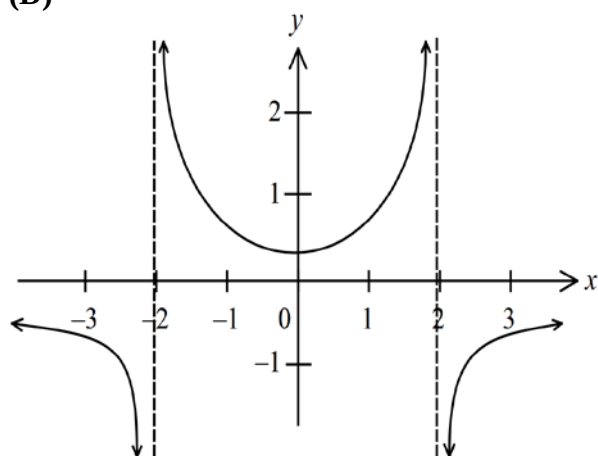
(B)



(C)



(D)



6 What is $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \cos x \, dx$ equivalent to?

(A) $2 \int_0^{\frac{\pi}{2}} x \cos x \, dx$

(B) 0

(C) $\pi - 2$

(D) $\frac{\pi}{2} - 1$

7 Which expression is equal to $\frac{1}{(x+1)(x^2+4)}$?

(A) $\frac{1}{3(x+1)} - \frac{x+1}{3(x^2+4)}$

(B) $\frac{1}{5(x+1)} - \frac{x-1}{5(x^2+4)}$

(C) $\frac{1}{5(x+1)} + \frac{x-1}{5(x^2+4)}$

(D) $\frac{1}{x+1} - \frac{x+4}{x^2+4}$

8 Which expression is equal to $\int \cos^3 x \, dx$?

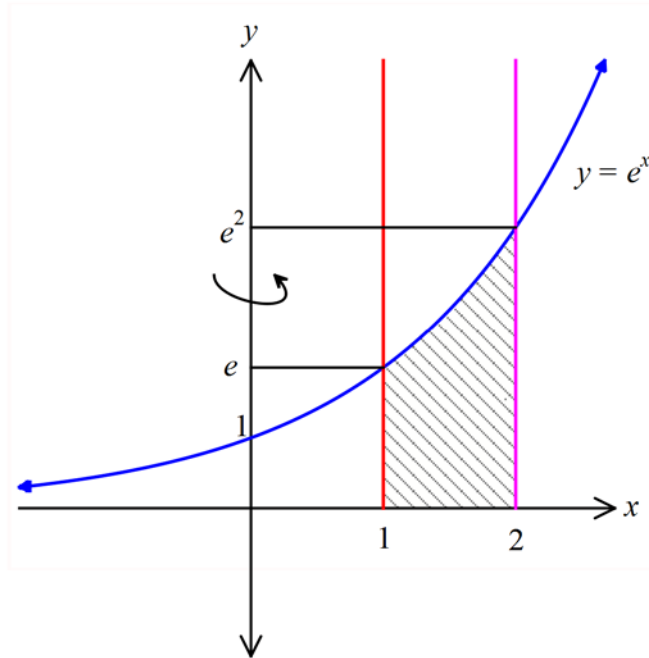
(A) $\frac{\cos^4 x}{4} + C$

(B) $3\cos^2 x + \sin x + C$

(C) $\sin x - \frac{\sin^3 x}{3} + C$

(D) $x - \frac{\sin^3 x}{3} + C$

- 9 The region bounded by the curve $y = e^x$, the x -axis, and the lines $x = 1$ and $x = 2$, is rotated around the y -axis to form a solid with volume V .



Which of the following is correct?

- (A) $V = \pi \int_e^{e^2} [4 - (\ln y)^2] dy.$
- (B) $V = 3\pi e + \pi \int_e^{e^2} [4 - (\ln y)^2] dy.$
- (C) $V = 3\pi e + \pi \int_e^{e^2} (4 - 2 \ln y) dx.$
- (D) $V = \pi \int_1^2 e^{2x} dx.$

$$J = \int_0^1 \sqrt{1-x^4} \, dx$$

$$K = \int_0^1 \sqrt{1+x^4} \, dx$$

$$L = \int_0^1 \sqrt{1-x^8} \, dx$$

Which of the following is true for the definite integrals shown above?

- (A) $J < L < 1 < K$
- (B) $J < L < K < 1$
- (C) $L < J < 1 < K$
- (D) $L < J < K < 1$

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section.

Answer each question in the appropriate writing booklet. Extra booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks). Use a Separate Booklet.

Marks

-
- (a) Write the complex number $\left(\frac{4i^7 - i}{1 + 2i}\right)^2$ in the form $a + bi$ where a and b are real numbers. 2

- (b) Using integration by parts, find $\int_0^1 x e^{2x} dx$. 2

- (c) Find the value of $\frac{dy}{dx}$ at the point $(1, 4)$ on the curve $4x^3 + xy^2 = 5xy$. 3

- (d) If $z = 4 + 2i$ and $w = -1 + 3i$, find $\arg(zw)$. 2

Question 11 continues on page 8.

Question 11 (continued).**Marks**

- (e) On an Argand diagram shade the region that is satisfied by both the conditions. 2

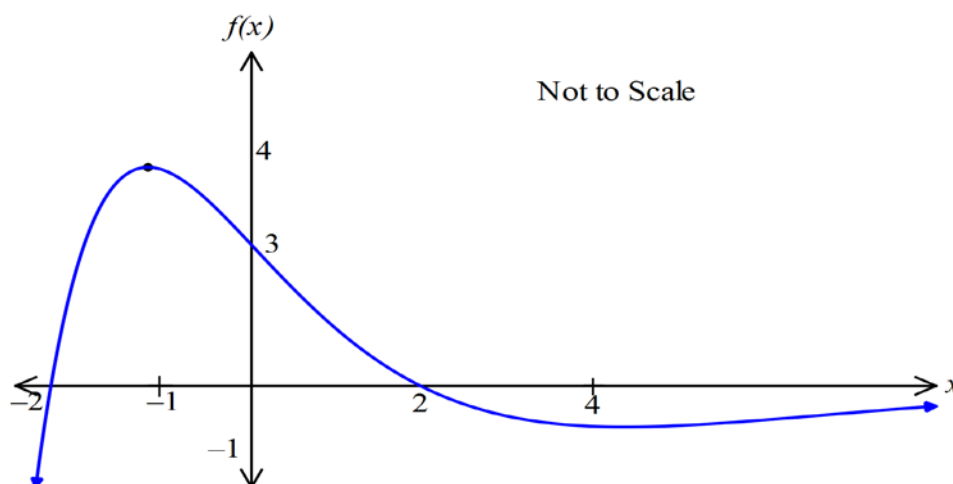
$$0 \leq \text{Arg}(z+i) \leq \frac{\pi}{4} \text{ and } |z-1| \leq \sqrt{2}.$$

- (f) If α, β and γ are roots of the equation $x^3 + 6x + 1 = 0$, find the polynomial equation whose roots are $\alpha\beta, \beta\gamma$ and $\alpha\gamma$. 2

- (g) Find $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin x \cos^3 x \, dx$. 2

End of Question 11

- (a) Given the sketch of the function $f(x)$, draw a one-third page sketch of each of the following on separate diagrams.



- | | |
|--|---|
| (i) $y = -f(x)$. | 1 |
| (ii) $y = \sqrt{f(x)}$. | 1 |
| (iii) $y = f(x^2)$. | 2 |
| | |
| (b) If $2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$ has a root of multiplicity of 3, factorise $2x^4 + 9x^3 + 6x^2 - 20x - 24$ fully. | 2 |

Question 12 continues on page 10.

Question 12 (continued).**Marks**

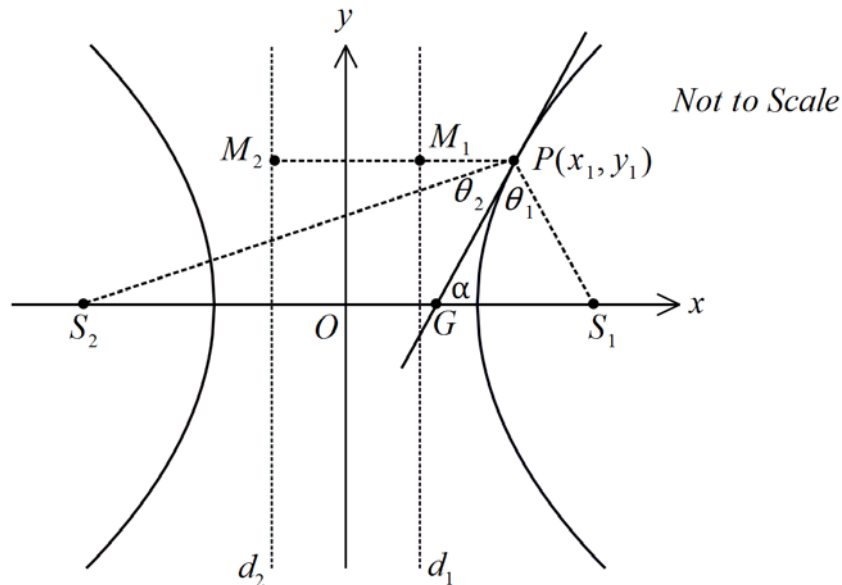
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- (c) The roots of the equation $z^3 + 2z^2 + 3z - 4 = 0$ are α , β and γ .
- (i) (1) Write down the value of $\alpha + \beta + \gamma$ and the value of $\alpha\beta + \beta\gamma + \gamma\alpha$. **1**
- (2) Hence show that $\alpha^2 + \beta^2 + \gamma^2 = -2$. **1**
- (ii) Find the value of $(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha)$. **2**
- (iii) Find a cubic equation whose roots are $\alpha + \beta$, $\beta + \gamma$ and $\gamma + \alpha$. **2**
- (d) Using the substitution $n = 1 - x$, evaluate $\int_{-1}^0 \frac{x}{(1-x)^4} dx$. **3**

End of Question 12

- (a) The area enclosed by the curve $y = (x-3)^2$ and the line $y = 9$ is rotated about the y -axis. Use the method of cylindrical shells to find the exact volume of the solid formed.

4

(b)



The point $P(x_1, y_1)$ lies on the hyperbola $\frac{x^2}{25} - \frac{y^2}{9} = 1$.

The two foci of the hyperbola are S_1 and S_2 and the two directrices are d_1 and d_2 , as shown.

- (i) Show that the length $S_1P = \frac{\sqrt{34}}{5} x_1 - 5$. 2
- (ii) Show that the equation of the tangent at P is $\frac{x_1 x}{25} - \frac{y_1 y}{9} = 1$. 2
- (iii) The tangent at P intersects the transverse axis at point G .
Find the coordinates of point G . 1
- (iv) Given $\angle S_1PG = \theta_1$, $\angle GPS_2 = \theta_2$ and $\angle S_1GP = \alpha$,
 - (1) By using the sine rule, show that $\sin \alpha = \frac{x_1 \sin \theta_1}{5}$. 2
 - (2) Hence, show that $\sin \theta_1 = \sin \theta_2$. 2
 - (3) Hence, deduce that GP bisects $\angle S_1PS_2$. 2

End of Question 13

Question 14 (15 marks). Use a Separate Booklet.

Marks

(a) Find all real x such that $3\sqrt{x(1-x)} < |x-2|$. **3**

(b) (i) Show that $x^4 + y^4 \geq 2x^2y^2$. **1**

(ii) If $P(x, y)$ is any point on the curve $x^4 + y^4 = 1$, prove that $OP \leq 2^{\frac{1}{4}}$ where O is the origin. **3**

(c) (i) (1) Use De Moivre's Theorem to show that if $z = \cos \theta + i \sin \theta$, then **1**

$$z^n - \frac{1}{z^n} = 2i \sin n\theta.$$

(2) Write down a similar expression for $z^n + \frac{1}{z^n}$. **1**

(ii) (1) Expand $\left(z - \frac{1}{z}\right)^2 \left(z + \frac{1}{z}\right)^2$ in terms of z . **1**

(2) Hence, show that $8\sin^2 \theta \cos^2 \theta = A + B \cos 4\theta$, where A and B are integers. **2**

(iii) Hence, by means of the substitution $x = 2 \sin \theta$, find the exact value of **3**

$$\int_1^2 x^2 \sqrt{4-x^2} \, dx.$$

End of Question 14

Question 15 (15 marks). Use a Separate Booklet.**Marks**

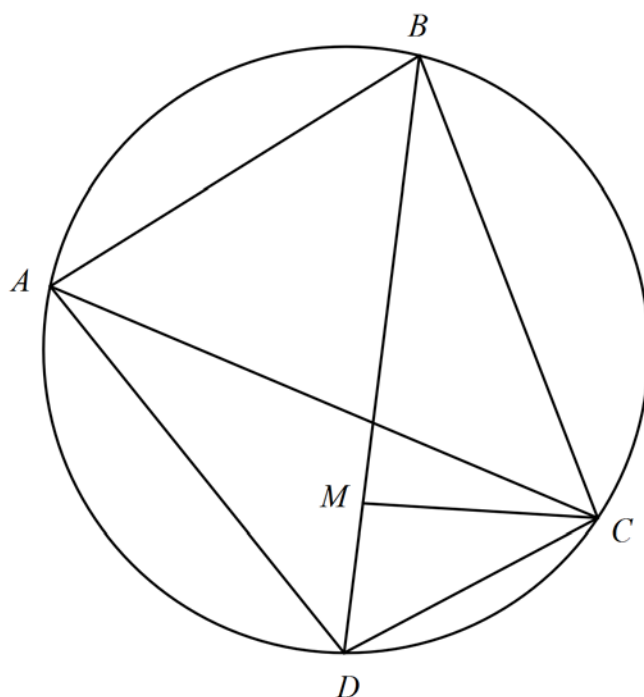
- (a) Draw a half page sketch of $y = \log_e |\tan x|$ for the domain $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. 2
- (b) (i) Use the factor theorem to show that $2x - 1$ is a factor of $8x^3 - 4x + 1$. 1
- (ii) Show that $4\cos 2\theta \cos \theta + 1$ can be written as $8x^3 - 4x + 1$ where $x = \cos \theta$. 1
- (iii) Given that $\theta = 72^\circ$ is a solution of $4\cos 2\theta \cos \theta + 1 = 0$, use the results from parts (i) and (ii) to show that the exact value of $\cos 72^\circ$ is $\frac{(\sqrt{5}-1)}{p}$ where p is a constant. 3
- (c) (i) Express $(k+1)^2 + 5(k+1) + 8$ in the form $k^2 + ak + b$, where a and b are constants. 1
- (ii) Prove by induction that, for all integers $n \geq 1$, 3
- $$\sum_{r=1}^n r(r+1)\left(\frac{1}{2}\right)^{r-1} = 16 - (n^2 + 5n + 8)\left(\frac{1}{2}\right)^{n-1}.$$
- (d) Given that $I_n = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot^n x \, dx$, for $n = 1, 2, \dots$
- (i) Show that $I_1 = \frac{1}{2} \ln 2$. 1
- (ii) Show that $I_{n-2} + I_n = \frac{1}{n-1} \left(3^{\frac{n-1}{2}} - 1 \right)$ for $n = 3, 4, 5, \dots$ 3

End of Question 15

- (a) The equation $z^4 + 4iz^3 - 4iz^2 + (8 + 8i)z - 32(1+i) = 0$ has roots $\alpha, \beta, -2\alpha, \gamma$ which represent the vertices A, B, C and D of a parallelogram in the Argand plane.
- (i) Using the properties of a parallelogram, show that $\alpha + \beta + \gamma = 0$. 2
- (ii) Hence, or otherwise, show that $\alpha = 2i$. 1
- (iii) Given $\sqrt{3 + 4i} = \pm(2 + i)$, find the vertices of the parallelogram $ABCD$. 3
- (b) (i) Show that $1 + \cos 2\theta + i \sin 2\theta = 2 \cos \theta (\cos \theta + i \sin \theta)$. 1
- (ii) Hence, prove that $\left(1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right)^n = -2^n \cos^n \frac{\pi}{n}$ where n is any integer. 1
- (iii) Hence, simplify $\left(1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right)^n - \left(1 + \cos \frac{2\pi}{n} - i \sin \frac{2\pi}{n}\right)^n$. 2

Question 16 continues on page 15

- (c) (i) Ptolemy's Theorem states that in a cyclic quadrilateral the product of the diagonals is equal to the sum of the products of the pairs of opposite sides, i.e. $AC \times BD = AB \times CD + BC \times AD$.
 M is the point on BD such that $\angle ACB = \angle DCM$.
Prove Ptolemy's theorem. 3



- (ii) Hence, if $AB = AD$, $\angle BCD$ is a right angle and the area of the quadrilateral $ABCD$ is 18 cm^2 , find the length of AC . 2

End of Paper

Extension 2 2016 Trial solutions

- | | |
|------|-------|
| 1. D | 6. B |
| 2. C | 7. B |
| 3. C | 8. C |
| 4. A | 9. B |
| 5. A | 10. A |

Question 11

$$a) \left(\frac{4i^4 - i}{1 + 2i} \right)^2$$

$$= \left(\frac{-5i}{1 + 2i} \right)^2$$

$$= \left[\frac{-5i(1 - 2i)}{1 + 4} \right]^2$$

$$= [-i(1 - 2i)]^2$$

$$= (-2 - i)^2$$

$$= 4 - 1 + 4i$$

$$= 3 + 4i$$

① → For getting to a point where realisation can happen

① Answer

$$b) \int_0^1 x e^{2x} dx$$

$$= \left[\frac{1}{2} x e^{2x} \right]_0^1 - \frac{1}{2} \int_0^1 e^{2x} dx$$

$$= \left(\frac{1}{2} e^2 - 0 \right) - \frac{1}{2} \left[\frac{1}{2} e^{2x} \right]_0^1$$

$$= \frac{1}{2} e^2 - \frac{1}{4} e^2 + \frac{1}{4} e^0$$

$$= \frac{1}{4} e^2 + \frac{1}{4}$$

① Answer

$$a) 4x^3 + xy^2 = 5xy$$

$$① \quad 12x^2 + 1 \cdot y^2 + 2y \cdot x \cdot \frac{dy}{dx} = 5 \cdot y + 5x \cdot 1 \cdot \frac{dy}{dx}$$

$$12x^2 + y^2 - 5y = (5x - 2xy) \frac{dy}{dx}$$

$$① \quad \frac{dy}{dx} = \frac{12x^2 + y^2 - 5y}{5x - 2xy}$$

When $x = 1$, $y = 4$:

$$\frac{dy}{dx} = \frac{12 + 4^2 - 5(4)}{5 - 2(1)(4)}$$

$$① \quad = \frac{-8}{3}$$

d) $z = 4 + 2i$ $w = -1 + 3i$

$zw = (4 + 2i)(-1 + 3i)$

$= -4 - 6 - 2i + 12i$

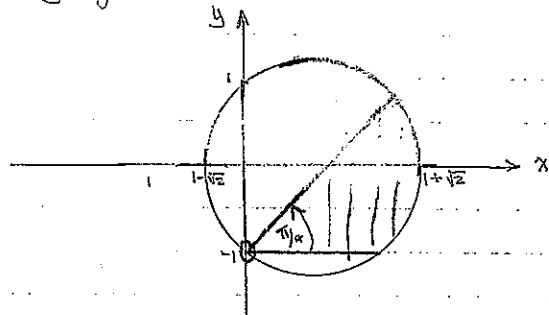
$= -10 + 10i$

$\arg zw = \frac{3\pi}{4}$

— ① For product

— ① For argument

e)



① for correctly placed circle

① Arc with open circle

Equation with roots α/β , β/γ and α/γ correctly

$= \frac{\alpha/\beta}{\gamma}$, $\frac{\beta/\gamma}{\alpha}$ and $\frac{\alpha/\gamma}{\beta}$ placed

$= \frac{-1}{\gamma}$, $\frac{-1}{\alpha}$ and $\frac{-1}{\beta}$

$(\frac{-1}{x})^3 + 6(\frac{-1}{x}) + 1 = 0$

$\frac{-1}{x^3} - \frac{6}{x} + 1 = 0$

① - Appropriate method

$-1 - 6x^2 + x^3 = 0$

$x^3 - 6x^2 - 1 = 0$ ① correct eqⁿ

g) $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin x \cos^3 x \, dx$

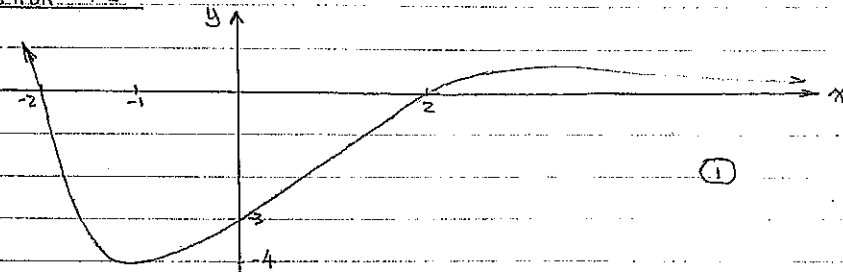
$= \frac{-1}{4} \cos^4 x \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$ — ① correct integral

$= -0 + \frac{1}{4} (\frac{1}{\sqrt{2}})^4$

$= \frac{1}{16}$ — ① correct answer

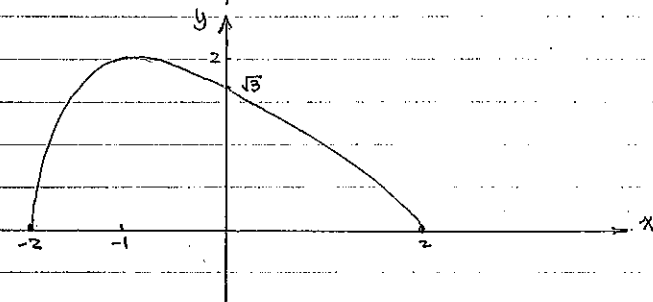
Question 12

a) i)



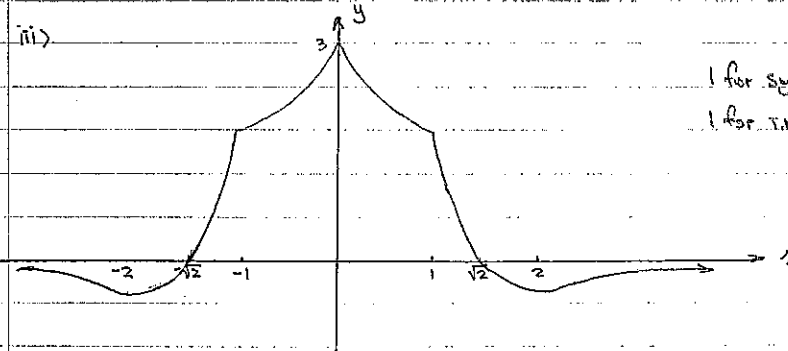
①

ii)



①

iii)



1 for symmetry
1 for intercepts

②

b) Let $P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$

$P'(x) = 8x^3 + 27x^2 + 12x - 20$

$P''(x) = 24x^2 + 54x + 12 = 0$

$4x^2 + 9x + 2 = 0$

$(4x + 1)(x + 2) = 0$

$x = -2$ or $x = -\frac{1}{4}$

$P(-\frac{1}{4}) \neq 0$ $P(-2) = 0$

$P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$

②

$$c) z^3 + 2z^2 + 3z - 4 = 0$$

$$i) \alpha + \beta + \gamma = -2$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = 3$$

$$\alpha^2 + \beta^2 + \gamma^2$$

$$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= (-2)^2 - 2(3)$$

$$= -2$$

①

$$ii) (\alpha + \beta)(\beta + \gamma)(\gamma + \alpha)$$

$$= 2\alpha\beta\gamma + \beta^2\gamma + \alpha\gamma^2 + \beta\gamma^2 + \alpha^2\beta + \beta^2\alpha + \alpha^2\gamma$$

$$= 2\alpha\beta\gamma + \alpha\beta(\alpha + \beta) + \alpha\gamma(\alpha + \gamma) + \beta\gamma(\beta + \gamma)$$

$$= 2\alpha\beta\gamma + \alpha\beta(-2 - \gamma) + \alpha\gamma(-2 - \beta) + \beta\gamma(-2 - \alpha)$$

$$= 2\alpha\beta\gamma - 2(\alpha\beta + \beta\gamma + \alpha\gamma) - 3\alpha\beta\gamma$$

$$= -2(3) - 1(4)$$

$$= -10$$

②

iii) Equation with roots $\alpha + \beta$, $\beta + \gamma$ and $\alpha + \gamma$

$$= \alpha + \beta + \gamma - \gamma, \alpha + \beta + \gamma - \alpha$$

$$\text{and } \alpha + \beta + \gamma - \beta$$

$$= -2 - \alpha, -2 - \beta \text{ and } -2 - \gamma$$

$$(-2 - \gamma)^3 + 2(-2 - \gamma)^2 + 3(-2 - \gamma) - 4 = 0$$

$$-(z^3 + 6z^2 + 12z + 8) + 2(z^2 + 4z + 4)$$

$$-6 - 3z - 4 = 0$$

$$z^3 + 4z^2 + 7z + 10 = 0$$

③

$$d) \int_{-1}^0 \frac{x}{(1-x)^4} dx$$

$$n = 1 - x \Rightarrow x = 1 - n$$

$$dn = -dx$$

$$= \int_2^1 \frac{n-1}{n^4} dn$$

1

$$x=0, n=1-0=1$$

$$x=-1, n=1-(-1)=2$$

$$= \int_2^1 \left(n^{-3} - n^{-4} \right) dn$$

$$= \left[-\frac{1}{2} n^{-2} + \frac{1}{3} n^{-3} \right]_2^1$$

$$= \left(-\frac{1}{2} + \frac{1}{3} \right) + \frac{1}{2} \times \frac{1}{4} - \frac{1}{3} \times \frac{1}{8}$$

$$= -\frac{1}{12}$$

③

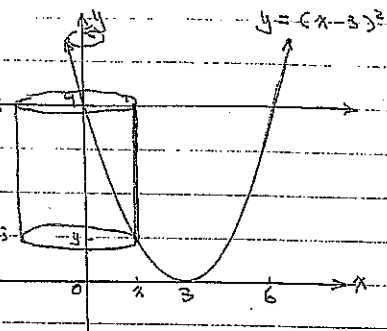
Q13

$$a) \delta V = \pi (x + \delta x)^2 (9 - y) - \pi x^2 (9 - y)$$

$$= \pi (9 - y) [(x + \delta x)^2 - x^2]$$

$$= \pi (9 - y) (2x + \delta x) \delta x$$

$$① V \approx \sum_{x=0}^6 \pi [9 - (x-3)^2] 2x \delta x$$

ignore δx^2 

$$① V = 2\pi \int_0^6 x^2 (6 - x) dx$$

$$= 2\pi \int_0^6 (6x^2 - x^3) dx$$

$$= 2\pi [2x^3 - \frac{1}{4}x^4]_0^6$$

$$= 2\pi [2 \times 6^3 - \frac{1}{4} \times 6^4 - 0]$$

$$① = 216\pi \text{ cubic units}$$

$$dV = 2\pi x (9 - y) dx$$

$$= 2\pi x (9 - (x-3)^2) dx$$

$$= 2\pi x (9 - (x^2 - 6x + 9)) dx$$

$$= 2\pi x (6x - x^2) dx$$

$$= 2\pi (6x^2 - x^3) dx$$

$$b) \frac{x^2}{25} - \frac{y^2}{9} = 1$$

$$ie = \sqrt{\frac{9}{25} + 1}$$

$$= \frac{\sqrt{34}}{5}$$

$$M_1 = (5 \times \frac{5}{\sqrt{34}}, y_1) = (\frac{25}{\sqrt{34}}, y_1)$$

$$PS_1 = e$$

$$PM_1$$

$$PS_1 = e \times PM_1$$

$$= \frac{\sqrt{34}}{5} (x_1 - \frac{25}{\sqrt{34}})$$

$$= \frac{\sqrt{34}}{5} x_1 - \frac{\sqrt{34}}{5} \times \frac{25}{\sqrt{34}}$$

$$= \frac{\sqrt{34}}{5} x_1 - 5$$

① correctly finding e and the x coordinate of M_1 or the eqⁿ of d_1

① correct demonstration

$$ii) \frac{2x}{25} - \frac{2y}{9} \times \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{2x}{25} \times \frac{9}{2y}$$

$$= \frac{9x}{25y}$$

$$m_P = \frac{9x_1}{25y_1}$$

Equation of tangent at P

$$\Rightarrow \frac{9x_1}{25y_1} = \frac{y - y_1}{x - x_1}$$

$$9x_1x - 9x_1^2 = 25y_1y - 25y_1^2$$

① Derivative

① obtaining eqⁿ

using a correct method

$$\frac{9x_1x}{9 \times 25} - \frac{25y_1y}{9 \times 25} = \frac{9x_1^2}{9 \times 25} - \frac{25y_1^2}{9 \times 25}$$

$$\frac{x_1x}{25} - \frac{y_1y}{9} = \frac{x_1^2}{25} - \frac{y_1^2}{9}$$

$$\frac{x_1x}{25} - \frac{y_1y}{9} = 1; \text{ since } P(x_1, y_1) \text{ lies on the hyperbola } \frac{x^2}{25} - \frac{y^2}{9} = 1$$

$$\text{so } \frac{x_1^2}{25} - \frac{y_1^2}{9} = 1$$

iii) When $y = 0$,

$$\frac{x_1x}{25} - 0 = 1$$

$$x_1x = 25$$

$$x = \frac{25}{x_1}$$

$$G = (\frac{25}{x_1}, 0)$$

1/r/w

$$iv) (1) GS_1 = \sqrt{34} - \frac{25}{x_1}$$

In $\triangle G S_1 P$,

$$\frac{PS_1}{\sin \angle S_1 G P} = \frac{GS_1}{\sin \angle S_1 P G}$$

$$\frac{\frac{\sqrt{34}}{5} x_1 - 5}{\sin \alpha} = \frac{\sqrt{34} - \frac{25}{x_1}}{\sin \theta_1}$$

① correct substitution

$$\sin \alpha = \frac{(\frac{\sqrt{34}}{5} x_1 - 5) \sin \theta_1}{\sqrt{34} - \frac{25}{x_1}}$$

of relevant

information

$$= \frac{\sqrt{34} x_1 - 25}{\sqrt{34} x_1 - \frac{25}{x_1}} \sin \theta_1$$

into sine rule

$$= \frac{\sqrt{34} x_1 - 25}{\sqrt{34} x_1 - \frac{25}{x_1}} \sin \theta_1$$

$$= \frac{x_1 \sin \theta_1}{5}$$

① correct manipulation to achieve the show.

$$(2) M_2 = (\frac{-25}{\sqrt{34}}, y_1)$$

$$PS_2 = e$$

$$PM_2$$

$$PS_2 = \frac{\sqrt{34}}{5} (x_1 + \frac{25}{\sqrt{34}})$$

$$= \frac{\sqrt{34}}{5} x_1 + 5$$

$$GS_2 = \frac{25}{x_1} + \sqrt{34}$$

In $\Delta S_2 P G$,

$$\frac{GS_2}{\sin \angle GPS_2} = \frac{PS_2}{\sin \angle S_2 P G}$$

$$\frac{\frac{25}{\sqrt{34}} + \sqrt{34}}{\sin \theta_2} = \frac{\frac{\sqrt{34}}{5} \times 1 + 5}{\sin (180^\circ - \alpha)}$$

$$\sin (180^\circ - \alpha) = \frac{\frac{\sqrt{34}}{5} \times 1 + 25}{\frac{25 + \sqrt{34} \times 1}{\sqrt{34}}} \times \sin \theta_2$$

$$\sin \alpha = \frac{\sqrt{34}}{5} \sin \theta_2 \quad \text{--- *}$$

$$\frac{\sqrt{34}}{5} \sin \theta_1 = \frac{\sqrt{34}}{5} \sin \theta_2 \quad ; \text{equating (1) and *}$$

$$\sin \theta_1 = \sin \theta_2$$

(3) From (2), $\sin \theta_1 = \sin \theta_2$,
 then $\theta_1 = \theta_2$ OR $\theta_1 = 180^\circ - \theta_2$.
 If $\theta_1 = 180^\circ - \theta_2$,
 then $\angle GPS_2 + \angle S_1 P G = \angle S_1 P S_2$
 $= \theta_2 + \theta_1$
 $= \theta_2 + 180^\circ - \theta_2$
 $= 180^\circ$

$\angle S_1 P S_2 = 180^\circ$ iff P lies on the x-axis.
 But if P lies on the x-axis, GP cannot
 bisect $\angle S_1 P S_2$ since they all lie on the x-axis.
 So $\theta_1 \neq 180^\circ - \theta_2$,
 $\theta_1 = \theta_2 \Rightarrow \angle S_1 P G = \angle GPS_2$,
 $\therefore$ GP bisects $\angle S_1 P S_2$.

(2) correct argument

(1) Partially correct

(2) correct argument

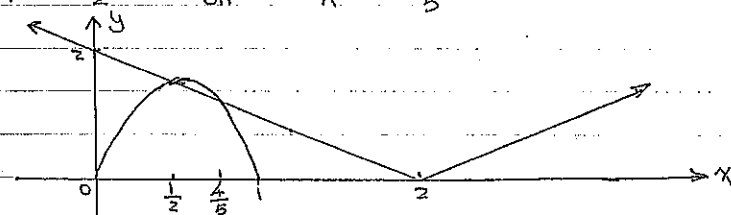
(1) Heading toward correct argument

Q14

$$3\sqrt{x(1-x)} < |x-2|$$

$$3\sqrt{x(1-x)} < \sqrt{(x-2)^2}$$

If $9x(1-x) = x^2 - 4x + 4$
 $9x - 9x^2 = x^2 - 4x + 4$
 $10x^2 - 13x + 4 = 0$
 $(2x-1)(5x-4) = 0$
 then $x = \frac{1}{2}$ OR $x = \frac{4}{5}$

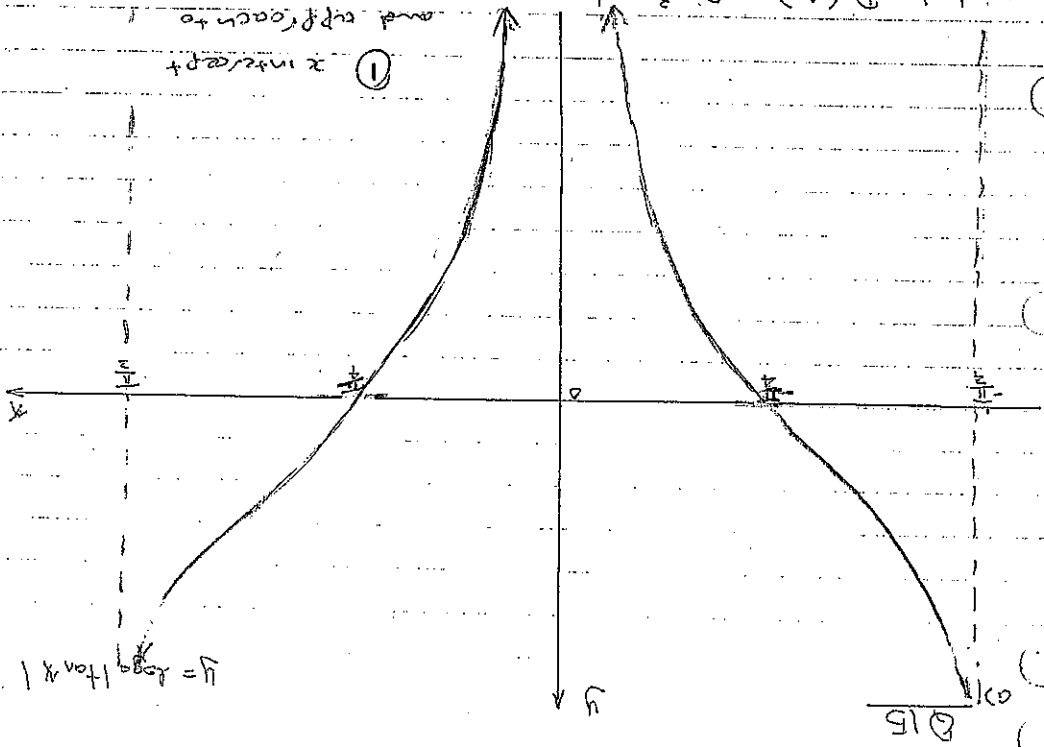


$$0 < x < \frac{1}{2} \text{ OR } \frac{4}{5} < x < 1$$

b) i) $(x^2 - y^2)^2 \geq 0$
 $x^4 - 2x^2y^2 + y^4 \geq 0$
 $x^4 + y^4 \geq 2x^2y^2$

ii) $OP = \sqrt{x^2 + y^2}$
 $OP^4 = (x^2 + y^2)^2$
 $= x^4 + y^4 + 2x^2y^2$
 $\leq x^4 + y^4 + x^4 + y^4$
 $= 1 + 1 = 2$
 $OP^4 \leq 2$
 $OP \leq \sqrt[4]{2}$

c) i) (1) $z = \cos \theta + i \sin \theta$
 $z^n = \cos n\theta + i \sin n\theta$
 $z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$
 $= \cos n\theta - i \sin n\theta$



Since $\theta = 72^\circ$ is a solution of $4 \cos 2\theta \cos \theta + 1 = 0$,
 $\chi = \cos 72^\circ$ is a solution of $P(\chi) = 8\chi^3 - 4\chi + 1$
 However $\cos 72^\circ \neq \frac{1}{2}$ and $\cos 72^\circ > 0$,
 so $\cos 72^\circ = \frac{1}{4}(-1 + \sqrt{5})$

$$\begin{aligned} \text{a) i) } (k+1)^2 + 5(k+1) + 8 \\ = k^2 + 2k + 1 + 5k + 5 + 8 \quad | \text{ r/w } \\ = k^2 + 7k + 14 \end{aligned}$$

$$\text{ii) let the statement be } \sum_{r=1}^n r(r+1)\left(\frac{1}{2}\right)^{r-1} = 16 - (n^2 + 5n + 8)\left(\frac{1}{2}\right)^{n-1}$$

Step 1: Show that the statement is true for $n=1$

$$\text{LHS} = 1(1+1)\left(\frac{1}{2}\right)^{1-1}$$

$$\begin{aligned} \text{① - correct setup} &= 2 \times \frac{1}{2} \\ \text{showing link to part i)} &= 2 \end{aligned}$$

$$\begin{aligned} \text{② - Body correct} \quad \text{RHS} &= 16 - (1^2 + 5 \times 1 + 8)\left(\frac{1}{2}\right)^{1-1} \\ \text{w.m conclusion} &= 16 - 14 \\ &= 2 \\ &= \text{LHS} \end{aligned}$$

Step 2: Assume statement is true for $n=k$;

$$\text{i.e. assume } \sum_{r=1}^k r(r+1)\left(\frac{1}{2}\right)^{r-1} = 16 - (k^2 + 5k + 8)\left(\frac{1}{2}\right)^{k-1}$$

Step 3: Prove statement is true for $n=k+1$;

$$\text{i.e. prove } \sum_{r=1}^{k+1} r(r+1)\left(\frac{1}{2}\right)^{r-1} = 16 - [(k+1)^2 + 5(k+1) + 8]\left(\frac{1}{2}\right)^k$$

$$\begin{aligned} \text{LHS} &= \sum_{r=1}^{k+1} r(r+1)\left(\frac{1}{2}\right)^{r-1} \\ &= 1(2)\left(\frac{1}{2}\right)^0 + 2(3)\left(\frac{1}{2}\right)^1 + \dots + k(k+1)\left(\frac{1}{2}\right)^{k-1} + (k+1)(k+2)\left(\frac{1}{2}\right)^k \\ &= 16 - (k^2 + 5k + 8)\left(\frac{1}{2}\right)^{k-1} + (k+1)(k+2)\left(\frac{1}{2}\right)^k \\ &= \left(\frac{1}{2}\right)^k [16\left(\frac{1}{2}\right)^k - (k^2 + 5k + 8)\left(\frac{1}{2}\right)^{k-1} + (k+1)(k+2)] \\ &= \left(\frac{1}{2}\right)^k [16 \times 2^k - 2(k^2 + 5k + 8) + k^2 + 3k + 2] \\ &= \left(\frac{1}{2}\right)^k [16 \times 2^k - 2k^2 - 10k - 16 + k^2 + 3k + 2] \\ &= \left(\frac{1}{2}\right)^k (16 \times 2^k - k^2 - 7k - 14) \end{aligned}$$

$$\begin{aligned} &= \left(\frac{1}{2}\right)^k [16 \times 2^k - (k^2 + 7k + 14)] \\ &= \left(\frac{1}{2}\right)^k [16 \times 2^k - ((k+1)^2 + 5(k+1) + 8)] \quad ; \text{ from ci) } \\ &= 2^{-k} \times 16 \times 2^k - \left(\frac{1}{2}\right)^k [(k+1)^2 + 5(k+1) + 8] \\ &= 16 - [(k+1)^2 + 5(k+1) + 8]\left(\frac{1}{2}\right)^k \\ &= \text{RHS} \end{aligned}$$

Step 4:

$$\text{di) } I_n = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot^n x \, dx$$

$$\text{is } I_1 = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot x \, dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos x}{\sin x} \, dx$$

$$= \left[\ln(\sin x) \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= \ln\left(\sin \frac{\pi}{2}\right) - \ln\left(\sin \frac{\pi}{6}\right)$$

$$= \ln\left(\frac{1}{2}\right) - \ln\left(\frac{1}{2}\right)$$

$$= \ln 2^{-\frac{1}{2}} - \ln 2^{-1}$$

$$= -\frac{1}{2} \ln 2 + \ln 2$$

$$= \frac{1}{2} \ln 2$$

| r/w

$$\text{ii) } I_{n-2} + I_n$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot^{n-2} x \, dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot^n x \, dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (\cot^{n-2} x + \cot^n x) \, dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot^n x (1 + \cot^{-2} x) \, dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot^n x (1 + \tan^2 x) \, dx \quad \text{--- ①}$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \tan^n x \cdot \sec^2 x \, dx$$

$$= \frac{1}{1-n} \left[\tan^{1-n} x \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= \frac{1}{1-n} \left[\tan^{1-n} \frac{\pi}{2} - \tan^{1-n} \frac{\pi}{6} \right]$$

$$= \frac{1}{1-n} \left[1 - \left(\frac{1}{\sqrt{3}}\right)^{1-n} \right]$$

$$= \frac{1}{1-n} \left[1 - \sqrt{3}^{n-1} \right]$$

$$= \frac{1}{1-n} (\sqrt{3}^{n-1} - 1)$$

$$= \frac{1}{n-1} (3^{\frac{n-1}{2}} - 1)$$

--- ①

① follow through

Q16

$$a) z^4 + 4iz^3 - 4iz^2 + (8i+8)z - (32+32i) = 0$$

is Diagonals of a parallelogram bisect each other,
i.e. midpoint of AC = midpoint of BD

$$\frac{1}{2}(\alpha - 2\alpha) = \frac{1}{2}(\beta + \gamma) \quad \text{--- *}$$

$$-\alpha = \beta + \gamma \quad \text{--- *}$$

$$ii) \text{ Sum of roots } = \alpha + \beta - 2\alpha + \gamma = \frac{-4i}{1} \quad \text{--- (2)}$$

$$-\alpha + \beta + \gamma = -4i \quad \text{--- (1)}$$

$$-\alpha + (-\alpha) = -4i \text{ from *}$$

$$-2\alpha = -4i$$

$$\alpha = 2i$$

$$iii) \alpha = 2i$$

$$-2\alpha = -4i$$

$\Rightarrow (z-2i)$ is a factor

$\Rightarrow (z+4i)$ is a factor

So $(z-2i)(z+4i) = z^2 + 2iz + 8$ is a factor

$$z^4 + 4iz^3 - 4iz^2 + (8i+8)z - (32+32i) = 0$$

$$(z^2 + 2iz + 8)(z^2 + 2iz - (4+4i)) = 0$$

$$\text{Consider } z^2 + 2iz - (4+4i) = 0$$

$$z = \frac{-2i \pm \sqrt{(2i)^2 - 4(1)(-4-4i)}}{2(1)}$$

$$= \frac{-2i \pm \sqrt{12 + 16i}}{2}$$

$$= \frac{-2i \pm 2\sqrt{3+4i}}{2}$$

$$= -i \pm (2+i)$$

$$= 2 \text{ OR } -2-2i$$

$\therefore$ Vertices are $2i, -4i, 2$ and $-2-2i$.

(3)

$$b) i) \text{ RHS} = 2 \cos \theta (\cos \theta + i \sin \theta)$$

$$= 2 \cos^2 \theta + 2 \sin \theta \cos \theta i \quad \text{--- (1)}$$

$$= 1 + \cos 2\theta + i \sin 2\theta$$

$$= \text{LHS}$$

$$ii) \text{ LHS} = \left(1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right)^n$$

$$= \left(2 \cos \frac{\pi}{n}\right)^n \left(\cos \frac{\pi}{n} + i \sin \frac{\pi}{n}\right)^n \text{ --- from (i)}$$

$$= 2^n \cos^n \frac{\pi}{n} \left[\cos\left(\frac{\pi}{n}\right)(n) + i \sin\left(\frac{\pi}{n}\right)(n)\right] \quad \text{--- (1)}$$

$$= 2^n \cos^n \frac{\pi}{n} (\cos \pi + i \sin \pi)$$

$$= 2^n \cos^n \left(\frac{\pi}{n}\right) \cdot (-1 + i0)$$

$$= -2^n \cos^n \frac{\pi}{n}$$

$$= \text{RHS}$$

$$iii) \text{ From (i), similarly } 2 \cos \theta (\cos \theta - i \sin \theta) = 1 + \cos 2\theta - i \sin 2\theta$$

$$\left(1 + \cos \frac{2\pi}{n} - i \sin \frac{2\pi}{n}\right)^n$$

$$= \left(2 \cos \frac{\pi}{n}\right)^n (\cos \pi - i \sin \pi)$$

$$= -2^n \cos^n \frac{\pi}{n}$$

$$\left(1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right)^n - \left(1 + \cos \frac{2\pi}{n} - i \sin \frac{2\pi}{n}\right)^n$$

$$= -2^n \cos^n \frac{\pi}{n} - (-2^n \cos^n \frac{\pi}{n})$$

$$= -2^n \cos^n \frac{\pi}{n} + 2^n \cos^n \frac{\pi}{n}$$

$$= 0$$

(2)

c) i) In $\triangle ABC$ and $\triangle DMC$;

$$\angle ACB = \angle DCM \text{ (given)}$$

$$\angle BAC = \angle MDC \text{ (angles on the circumference standing on the same arc BC)}$$

$\triangle ABC \sim \triangle DMC$ (equiangular) 1.

$$\frac{AC}{CD} = \frac{AB}{MD} \text{ (corresponding sides of similar triangles)}$$

$$AB \times CD = AC \times MD$$

In $\triangle ACD$ and $\triangle BCM$;

$$\begin{aligned} \angle ACD &= \angle ACH + \angle DCM \text{ (adjacent angles)} \\ &= \angle ACH + \angle ACB \text{ (}\angle ACB = \angle DCM \text{, given)} \\ &= \angle BCM \text{ (adjacent angles)} \end{aligned}$$

$$\angle DAC = \angle MBC \text{ (angles on the circumference standing on the same arc CD)}$$

$\triangle ACD \sim \triangle BCM$ (equiangular) 1

$$\frac{AD}{BM} = \frac{AC}{BC} \text{ (corresponding sides of similar triangles)}$$

$$AD \times BC = AC \times BM$$

$$\text{So } AB \times CD + BC \times AD$$

$$= AC \times MD + AC \times BM$$

$$= AC (MD + BM)$$

$$= AC \times BD$$

ii) $\angle BAD = \angle BCD$ ($\angle BCD = 90^\circ$ given and opposite angles of cyclic quadrilateral are supplementary)

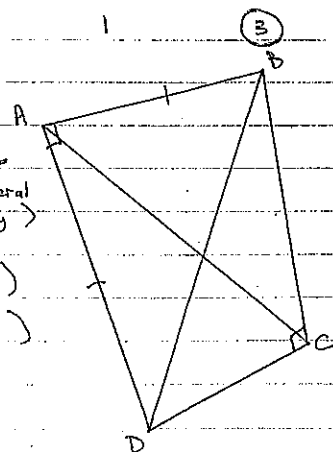
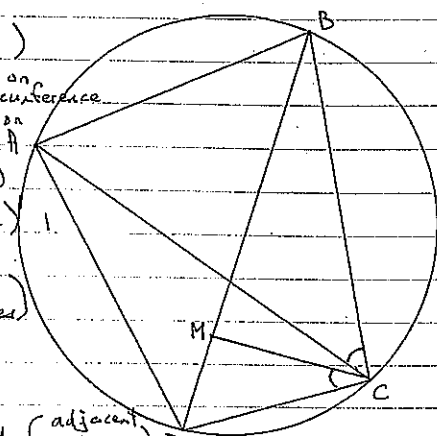
In $\triangle ABD$;

$$\begin{aligned} BD^2 &= AB^2 + AD^2 \text{ (by Pythagoras')} \\ &= AB^2 + AB^2 \text{ (AD = AB, given)} \\ &= 2AB^2 \end{aligned}$$

$$BD = \sqrt{2} AB \quad \text{--- } \star$$

$$AC \times BD = AB \times CD + BC \times AD$$

$$\begin{aligned} \Rightarrow AC \times \sqrt{2} AB &= AB \times CD + BC \times AB \\ &= AB (CD + BC) \end{aligned}$$



$$AC = \frac{1}{\sqrt{2}} (CD + BC)$$

$$\begin{aligned} AC^2 &= \frac{1}{2} (CD + BC)^2 \\ &= \frac{1}{2} (CD^2 + BC^2 + 2(BC \times CD)) \\ &= \frac{1}{2} (2AB^2 + 2(BC \times CD)) \text{ ; from } \star \\ &= AB^2 + BC \times CD \quad \text{--- } \textcircled{1} \end{aligned}$$

$$\text{Area of } ABCD = \text{Area of } \triangle ABD + \text{Area of } \triangle BCD$$

$$18 = \frac{1}{2} (AB \times AD) + \frac{1}{2} (BC \times CD)$$

$$18 = \frac{1}{2} AB^2 + \frac{1}{2} (BC \times CD)$$

$$36 = AB^2 + (BC \times CD) \quad \text{--- } \textcircled{2}$$

$$\textcircled{1} \Rightarrow AC^2 = AB^2 + BC \times CD$$

$$\begin{aligned} &= AB^2 + 36 - AB^2 \text{ ; from } \textcircled{2} \\ &= 36 \end{aligned}$$

$$AC = 6 \text{ cm}$$

$\textcircled{2}$

* In $\triangle BCD$;

$$BD^2 = BC^2 + CD^2 \text{ (by Pythagoras')}$$

$$2AB^2 = BC^2 + CD^2 \text{ (from } \star \text{ where } BD = \sqrt{2} AB \text{)}$$